An Introduction to Number Theory and Modular Arithmetic

Ann Zhou

12 October 2026

1 Introduction

Instead of studying for my next NSML on Counting and Probability, a topic I have avoided since Freshman year, I am instead writing this document in an attempt to not only learn OverLeaf but delude myself into feeling productive while avoiding the only task I need to complete (cannon event). I hope you find this document helpful. Forgive me if you deem the organization of this document poor. Forever hold your peace because I probably won't fix it. Though, if there is anything conceptually wrong or any wrong answers, I will probably catch it soon. Trust your work, not mine.

2 Foundations of Number Theory

2.1 Prime and Composite Numbers

- **What it is:** A **prime number** is an integer $p > 1$ whose only positive divisors are 1 and p. A **composite number** is an integer $n > 1$ that has at least one positive divisor other than 1 and itself.
- **Equations/Definitions:**

$$\begin{aligned} p \text{ is prime} &\iff \text{if } p = ab \text{ for positive integers } a, b, \text{ then } \{a, b\} = \{1, p\} \\ n \text{ is composite} &\iff \exists\, a, b \in \mathbb{Z}^+ \text{ such that } n = ab \text{ and } 1 < a, b < n \end{aligned}$$

- **How/What it is used for:** Primes serve as the multiplicative building blocks for integers.
- **Tricks & Common Mistakes:**
  - **The Number 1:** 1 is **neither** prime nor composite.
  - **Parity:** 2 is the **only** even prime number. All other primes are odd.
  - **Primality Testing:** To check if n is prime, you only need to test divisibility by primes up to $\lfloor \sqrt{n} \rfloor$.
- **Example Question:** Find the only prime number p such that $p^2 + 2$ is also prime.
  Solution: If $p = 3$, $3^2 + 2 = 11$, which is prime. For any other prime, $p \equiv \pm 1 \pmod 3$, so $p^2 + 2 \equiv 1 + 2 \equiv 0 \pmod 3$, meaning $p^2 + 2$ will be a multiple of 3 greater than 3 (hence composite). Thus, **3** is the only solution.

2.2 Coprime Numbers

- **What it is:** Two integers a and b are **coprime** (or relatively prime) if they share no common positive factors other than 1.
- **Equation:** $\gcd(a, b) = 1$
- **How/What it is used for:** Essential for determining if a modular inverse exists and simplifying fractions.

- **Tricks & Common Mistakes:**
  - Numbers do not need to be individually prime to be coprime (e.g., $\gcd(8, 9) = 1$).
  - Consecutive integers are always coprime: $\gcd(n, n + 1) = 1$.
- **Example Question:** Find the number of integers x such that $1 \leq x \leq 20$ and $\gcd(x, 20) = 1$.
  Solution: The prime factors of 20 are 2 and 5. We remove multiples of 2 (10 numbers) and multiples of 5 (4 numbers), adding back the multiple of 10 (2 numbers) counted twice. $20 - 10 - 4 + 2 = \mathbf{8}$.

2.3 Prime Factorizations

- **What it is:** The unique way of expressing any integer $n > 1$ as a product of prime powers.
- **Equation (Canonical Form):**
$$n = p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k} = \prod_{i=1}^{k} p_i^{e_i}$$
- **How/What it is used for:** Finding the number of factors $\tau(n) = (e_1 + 1)(e_2 + 1) \cdots (e_k + 1)$ and the sum of factors.
- **Tricks & Common Mistakes:**
  - Remember to add 1 to the exponents when counting factors.
- **Example Question:** How many positive factors does 360 have?
  Solution: The prime factorization of 360 is $2^3 \cdot 3^2 \cdot 5^1$. The number of factors is $(3+1)(2+1)(1+1) = 4 \cdot 3 \cdot 2 = \mathbf{24}$.

2.4 Division and Remainders

- **What it is:** The Division Algorithm guarantees a unique quotient q and a non-negative remainder r.
- **Equation:** $a = qb + r$ where $q, r \in \mathbb{Z}$ and $0 \leq r < b$.
- **How/What it is used for:** Setting up algebraic equations and executing the Euclidean Algorithm.
- **Tricks & Common Mistakes:**
  - **Negative Dividends:** If $a = -5$ and $b = 3$, then $-5 = 3(-2) + 1$, so the remainder is 1, **not** -2. Remainder r must satisfy $0 \leq r < b$.
- **Example Question:** When a positive integer n is divided by 7, the remainder is 5. What is the remainder when $3n + 4$ is divided by 7?
  Solution: Write $n = 7q + 5$. Then $3n + 4 = 3(7q + 5) + 4 = 21q + 15 + 4 = 21q + 19$. We divide 19 by 7, which gives $19 = 7(2) + 5$. The remainder is $\mathbf{5}$.

2.5 Divisibility

- **What it is:** An integer a divides an integer b if there exists an integer k such that $b = ak$.
- **Equation:** $a \mid b \iff b = ak$ for some $k \in \mathbb{Z}$
- **How/What it is used for:** Setting up constraints. If $a \mid b$ and $b \neq 0$, then $|a| \leq |b|$.
- **Tricks & Common Mistakes:**
  - Linearity: If $d \mid a$ and $d \mid b$, then $d \mid (xa + yb)$ for any integers x, y.
  - 0 is divisible by every integer ($a \mid 0$ is true), but no integer is divisible by 0.

- **Example Question:** Find all positive integers n such that $n+2 \mid 3n+14$.
  Solution: By linearity, $n+2$ must also divide $(3n+14)-3(n+2) = 3n+14-3n-6 = 8$. The positive divisors of 8 are $1, 2, 4, 8$. Since n must be positive, $n+2 > 2$, leaving $n+2 = 4 \implies n = 2$ and $n+2 = 8 \implies n = 6$. The solutions are $n = \mathbf{2}, \mathbf{6}$.

3 Modular Arithmetic

3.1 Modular Congruence

- **What it is:** A statement that two numbers a and b leave the same remainder when divided by n.
- **Equation:** $a \equiv b \pmod{n} \iff n \mid (a-b)$
- **How/What it is used for:** Simplifying large arithmetic computations by replacing numbers with their remainders.
- **Tricks & Common Mistakes:**
  - **Division Trap:** If $ca \equiv cb \pmod{n}$, then $a \equiv b \pmod{\frac{n}{\gcd(c,n)}}$. You can only drop c directly if $\gcd(c, n) = 1$.
- **Example Question:** Find the remainder when $2023 \cdot 2024 \cdot 2025$ is divided by 11.
  Solution: Reduce each number modulo 11 first: $2023 \equiv 10 \pmod{11}$, $2024 \equiv 0 \pmod{11}$, and $2025 \equiv 1 \pmod{11}$. Thus, $10 \cdot 0 \cdot 1 \equiv \mathbf{0} \pmod{11}$.

3.2 Modular Inverses

- **What it is:** The modular equivalent of a reciprocal or fraction. For an integer a under a modulus m, its modular inverse is an integer x that satisfies the property that their product is congruent to 1 modulo m.
- **Equations/Definitions:** The modular inverse of a modulo m is denoted as a^{-1} or x, and is defined by the linear congruence:
  $$ax \equiv 1 \pmod{m}$$
  A unique solution x exists in the range $0 \leq x < m$ if and only if:
  $$\gcd(a, m) = 1$$
- **How/What it is used for:**
  - **Modular Division:** Solving algebraic expressions or fractions modulo m. Instead of dividing by a, you multiply by its modular inverse a^{-1}. For example, $\frac{b}{a} \equiv b \cdot a^{-1} \pmod{m}$.
  - **Linear Congruences:** Isolating variables in equations. A system or equation like $ax \equiv b \pmod{m}$ is solved directly by computing $x \equiv b \cdot a^{-1} \pmod{m}$.
- **Tricks & Common Mistakes:**
  - **The Existence Trap:** Never assume an inverse exists without checking coprimality. For example, $2x \equiv 1 \pmod{6}$ has no solution because $\gcd(2, 6) = 2 \neq 1$.
  - **The Negative Output:** When using algorithms like the Extended Euclidean Algorithm, your math might yield a negative value. Always normalize it by adding m until it falls into the valid range $[0, m-1]$.
- **Example Question:** Find the unique modular inverse of 11 modulo 37 such that $0 \leq x < 37$.
  Solution: Since 37 is prime and $\gcd(11, 37) = 1$, a unique inverse exists. We can compute it systematically using the Extended Euclidean Algorithm (running the Euclidean Algorithm backwards):

1. Perform forward division steps to track remainders:

$$37 = 3 \times 11 + 4 \implies 4 = 37 - 3 \times 11$$

$$11 = 2 \times 4 + 3 \implies 3 = 11 - 2 \times 4$$

$$4 = 1 \times 3 + 1 \implies 1 = 4 - 1 \times 3$$

2. Substitute backwards to express 1 as a linear combination of 11 and 37:

$$1 = 4 - 1 \times (11 - 2 \times 4) = 3 \times 4 - 1 \times 11$$

$$1 = 3 \times (37 - 3 \times 11) - 1 \times 11 = 3 \times 37 - 10 \times 11$$

3. Take the resulting equation modulo 37:

$$1 \equiv -10 \times 11 \pmod{37}$$

4. Clean up the negative scalar value to find the positive residue:

$$x \equiv -10 \equiv -10 + 37 = 27 \pmod{37}$$

The modular inverse is **27**. Check: $11 \times 27 = 297 = (37 \times 8) + 1 \equiv 1 \pmod{37}$.

3.3 Fermat's Little Theorem (FLT)

- **What it is:** A foundational result in number theory that dictates how integer powers behave under prime moduli, establishing a predictable cyclic period for modular exponentiation.
- **Equations/Definitions:** If p is a prime number and a is any integer such that $\gcd(a, p) = 1$, then:

$$a^{p-1} \equiv 1 \pmod{p}$$

  Alternatively, for any integer a without restriction on coprimality:

$$a^p \equiv a \pmod{p}$$

- **How/What it is used for:**
  - **Exponent Compression:** Reducing giant exponents by computing the power's exponent modulo $p - 1$ rather than modulo p.
  - **Modular Inverses:** Finding explicit formulas for modular fractions. Since $a \cdot a^{p-2} = a^{p-1} \equiv 1 \pmod{p}$, the modular inverse is given directly by:

$$a^{-1} \equiv a^{p-2} \pmod{p}$$

- **Tricks & Common Mistakes:**
  - **The Exponent Mistake:** Never reduce the exponent modulo p. You must always reduce it modulo $p - 1$. For example, evaluating $2^{11} \pmod{11}$ requires taking $11 \pmod{10} = 1$, which simplifies directly to $2^1 = 2$.
  - **Base Multiples Trap:** If a is a multiple of p, then $a^{p-1} \equiv 0 \pmod{p}$, not 1. Always verify that $\gcd(a, p) = 1$ before implementing the $a^{p-1} \equiv 1$ form.
- **Example Question:** Find the remainder when 3^{2026} is divided by 17.

  Solution: Since 17 is prime and $\gcd(3, 17) = 1$, we can apply Fermat's Little Theorem with $p - 1 = 16$.

  1. Divide the exponent 2026 by 16 to find the modular remainder:

$$2026 = 16 \times 126 + 10 \implies 2026 \equiv 10 \pmod{16}$$

2. Rewrite the original expression using the compressed exponent:

$$3^{2026} \equiv 3^{10} \pmod{17}$$

3. Evaluate $3^{10} \pmod{17}$ by repeatedly squaring small parts:

$$3^3 = 27 \equiv 10 \pmod{17}$$

$$3^9 = (3^3)^3 \equiv 10^3 = 1000 \pmod{17}$$

Since $1000 = 17 \times 58 + 14$, we have $3^9 \equiv 14 \equiv -3 \pmod{17}$.

$$3^{10} = 3 \cdot 3^9 \equiv 3 \cdot (-3) = -9 \equiv 8 \pmod{17}$$

The remainder is **8**.

3.4 Wilson's Theorem

- **What it is:** A structural assertion concerning the product of all non-zero remainders under a prime modulus, demonstrating that intermediate terms cleanly pair off into modular inverse pairs, leaving only self-inverses remaining.
- **Equations/Definitions:** An integer $p > 1$ is a prime number if and only if the factorial of $p - 1$ satisfies:

$$(p-1)! \equiv -1 \equiv p - 1 \pmod{p}$$

- **How/What it is used for:**
  - **Primality Testing:** It provides an exact, non-probabilistic algebraic condition to prove whether an arbitrary integer is prime or composite.
  - **Factorial Simplification:** Evaluating large factorial products modulo a prime by working backward from $(p-1)!$.
- **Tricks & Common Mistakes:**
  - **Composite Fallacy:** Wilson's Theorem does not hold for composite numbers. For any composite $n > 4$, $(n-1)! \equiv 0 \pmod{n}$ because the factors of n appear explicitly within the sequence of product terms.
  - **Backtracking Shift Signs:** When shifting Wilson's Theorem down to find smaller factorials near p, be careful with alternating negative signs. For instance, the first downward step shifts via:

$$(p-1)(p-2)! \equiv -1 \pmod{p} \implies (-1)(p-2)! \equiv -1 \pmod{p} \implies (p-2)! \equiv 1 \pmod{p}$$

- **Example Question:** Find the remainder when $19! \cdot 7!$ is divided by 23.

  Solution: We utilize Wilson's Theorem backwards to compute the value of the large factorial $19! \pmod{23}$:

1. Apply Wilson's Theorem for the prime $p = 23$:

$$22! \equiv -1 \pmod{23}$$

2. Expand the factorial terms to isolate 19!:

$$22 \cdot 21 \cdot 20 \cdot 19! \equiv -1 \pmod{23}$$

3. Substitute each coefficient with its equivalent negative remainder modulo 23:

$$(-1) \cdot (-2) \cdot (-3) \cdot 19! \equiv -1 \pmod{23}$$

$$-6 \cdot 19! \equiv -1 \pmod{23} \implies 6 \cdot 19! \equiv 1 \pmod{23}$$

4. Multiply both sides by the modular inverse of 6 modulo 23, which is 4 (since $6 \times 4 = 24 \equiv 1 \pmod{23}$):
$$19! \equiv 4 \pmod{23}$$
5. Compute the value of the smaller factorial $7! \pmod{23}$:
$$7! = 5040$$
Dividing 5040 by 23 gives a quotient of 219 with a remainder of 3 (since $23 \times 219 = 5037$). Thus, $7! \equiv 3 \pmod{23}$.
6. Combine the components to achieve the final product:
$$19! \cdot 7! \equiv 4 \cdot 3 = 12 \pmod{23}$$

The remainder is **12**.

3.5 Chinese Remainder Theorem (CRT)

- **What it is:** The Chinese Remainder Theorem (CRT) is a fundamental structural result in modular arithmetic. It states that if one knows the remainders of an unknown integer divided by several pairwise coprime integers, then the remainder of the division of the unknown integer by the product of these divisors can be uniquely determined. Essentially, it allows us to break down a large, complex modular congruence problem into a system of smaller, simpler independent congruences.
- **Equations/Definitions:** Let $m_1, m_2, \ldots, m_k$ be positive integers that are pairwise coprime (i.e., $\gcd(m_i, m_j) = 1$ for any $i \neq j$). Let $a_1, a_2, \ldots, a_k$ be any integers. The system of simultaneous linear congruences:
$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_k \pmod{m_k} \end{aligned}$$
has a unique solution for x modulo M, where $M = m_1 \cdot m_2 \cdots m_k$. To construct the unique solution $x \pmod{M}$:
  1. Compute the total product: $M = \prod_{i=1}^{k} m_i$.
  2. For each i, compute the partial product: $M_i = \frac{M}{m_i}$.
  3. For each i, compute the modular inverse: $M_i^{-1} \pmod{m_i}$, which is an integer M_i' such that $M_i M_i' \equiv 1 \pmod{m_i}$.
  4. The solution is formulated as: $x \equiv \sum_{i=1}^{k} a_i M_i M_i' \pmod{M}$.
- **Tricks & Common Mistakes:**
  - **Moduli Must Be Pairwise Coprime:** If $\gcd(m_i, m_j) \neq 1$, standard CRT cannot be applied blindly. The system might have no solution at all, or require an extended version of the theorem.
  - **Simplify Coefficients First:** Ensure the system is in the standard format $x \equiv a_i \pmod{m_i}$. If given $cx \equiv a_i \pmod{m_i}$, you must multiply by the modular inverse of c modulo m_i to isolate x first.
  - **Negative Remainders:** Modulo operations in standard programming languages can yield negative numbers. Always adjust intermediate values to be strictly positive: `(x % m + m) % m`.

- **Example Question:** Find the smallest positive integer x such that:

$$\begin{aligned} x &\equiv 2 \pmod 3 \\ x &\equiv 3 \pmod 5 \\ x &\equiv 2 \pmod 7 \end{aligned}$$

Solution: We follow the construction method:

1. Verify Coprimality: The moduli are $m_1 = 3, m_2 = 5, m_3 = 7$. They are pairwise coprime since $\gcd(3,5) = \gcd(5,7) = \gcd(3,7) = 1$.
2. Calculate M: $M = 3 \times 5 \times 7 = 105$.
3. Calculate M_i and Modular Inverses M_i':
   - For $m_1 = 3$: $M_1 = \frac{105}{3} = 35$. Find $35^{-1} \pmod 3 \implies 2^{-1} \pmod 3 \implies M_1' = 2$.
   - For $m_2 = 5$: $M_2 = \frac{105}{5} = 21$. Find $21^{-1} \pmod 5 \implies 1^{-1} \pmod 5 \implies M_2' = 1$.
   - For $m_3 = 7$: $M_3 = \frac{105}{7} = 15$. Find $15^{-1} \pmod 7 \implies 1^{-1} \pmod 7 \implies M_3' = 1$.
4. Assemble the final solution:

$$\begin{aligned} x &\equiv (a_1 M_1 M_1') + (a_2 M_2 M_2') + (a_3 M_3 M_3') \pmod M \\ x &\equiv (2 \cdot 35 \cdot 2) + (3 \cdot 21 \cdot 1) + (2 \cdot 15 \cdot 1) \pmod{105} \\ x &\equiv 140 + 63 + 30 = 233 \equiv 23 \pmod{105} \end{aligned}$$

The smallest positive integer solution is **23**.

3.6 Euler's Totient Function

- **What it is:** Euler's totient function, denoted as $\phi(n)$, counts the number of positive integers up to n that are relatively prime (coprime) to n.
- **Equations/Definitions:** If the prime factorization of n is $p_1^{e_1} p_2^{e_2} \cdots p_k^{e_k}$, then the product formula is:

$$\phi(n) = n \prod_{i=1}^{k} \left(1 - \frac{1}{p_i}\right) = p_1^{e_1 - 1}(p_1 - 1) \cdots p_k^{e_k - 1}(p_k - 1)$$

For a prime p, $\phi(p) = p - 1$. For two coprime integers a and b, $\phi(ab) = \phi(a)\phi(b)$.

- **How/What it is used for:** It computes the size of the multiplicative group of integers modulo n. It is primarily used to compress massive exponents via Euler's Totient Theorem: $a^{\phi(m)} \equiv 1 \pmod m$ when $\gcd(a, m) = 1$.
- **Tricks & Common Mistakes:**
  - **Prime Powers:** For a single prime power, remember it is $\phi(p^e) = p^e - p^{e-1}$, not $(p-1)^e$.
  - **Even Numbers:** For any $n > 2$, $\phi(n)$ is always an even integer.
- **Example Question:** Calculate $\phi(360)$.

Solution: The prime factorization of 360 is $2^3 \cdot 3^2 \cdot 5^1$. Applying the product formula:

$$\phi(360) = 360 \times \left(1 - \frac{1}{2}\right) \times \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{1}{5}\right)$$

$$\phi(360) = 360 \times \frac{1}{2} \times \frac{2}{3} \times \frac{4}{5} = 360 \times \frac{8}{30} = 12 \times 8 = 96.$$

4 Practice Problems

4.1 Foundational Problems

1. **(1992 AHSME, Problem 17)** The two-digit integers from 19 to 92 are written consecutively to form the large integer $N = 192021\ldots909192$. Suppose that 3^k is the highest power of 3 that is a factor of N. What is k?
Answer: $k = 3$
Not-So-Brief Hint: Use the Divisibility Rule of 3 and 9 (Modulo 3 and 9). A number is congruent to the sum of its digits modulo 9. Summing the individual digits of all numbers from 19 to 92 yields a sum that is a multiple of 27 but not 81, making $k = 3$.

2. **(2017 AMC 12A, Problem 11)** Modulo 10, what is the remainder of the sum of the first 45 terms of the sequence $1, 11, 111, 1111, \ldots$?
Answer: 5
Not-So-Brief Hint: Track the modulo 10 pattern. Each term $T_n \equiv 1 \pmod{10}$ for any positive integer index. Since we are adding 45 terms, the total sum is congruent to $45 \times 1 \equiv 5 \pmod{10}$.

3. **(2017 AMC 12A, Problem 18)** For some positive integer n, the number $n^2 + 2n + 3$ is a multiple of 5. What is the remainder when $(n+1)^2 + 2(n+1) + 3$ is divided by 5?
Answer: 4
Not-So-Brief Hint: Use polynomial substitution under prime modulus. Knowing $n^2 + 2n + 3 \equiv 0 \pmod{5}$, expanding the next expression gives $(n^2 + 2n + 1) + (2n + 2) + 3 = (n^2 + 2n + 3) + 2n + 3$. Testing the valid residues for n under mod 5 shows the final result cleanly reduces to 4.

4. **(2000 AMC 12, Problem 18)** In year N, the 300th day of the year is a Tuesday. In year $N + 1$, the 200th day is also a Tuesday. On what day of the week did the 100th day of the year $N - 1$ occur?
Answer: Thursday
Not-So-Brief Hint: There are either $65 + 200 = 265$ or $66 + 200 = 266$ days between the first two dates depending on whether year N is a leap year. Since 7 divides 266, year $N + 1$ is a leap year and $N - 1$ is not. Counting backwards from year N to year $N - 1$ leaves a remainder of 5, which puts you 5 days before Tuesday (landing on Thursday).

5. **(2021 AMC 10A, Problem 12)** What is the remainder when 2^{2021} is divided by 9?
Answer: 5
Not-So-Brief Hint: Apply Euler's Totient Function $\phi(m)$. Here, $m = 9$, so $\phi(9) = 9 \times (1 - \frac{1}{3}) = 6$. By Euler's Theorem, $2^6 \equiv 1 \pmod{9}$. Slicing the exponent: $2021 = 6 \times 336 + 5$, yielding $2^{2021} \equiv 1^{336} \times 2^5 \equiv 32 \equiv 5 \pmod{9}$.

4.2 Some Harder Problems

1. **(2017 AIME I, Problem 9)** Find the number of ordered pairs of integers (m, n) such that $1 \leq m \leq n \leq 1000$ and the least common multiple of m and n is a multiple of 3.
Answer: $667,167$
Not-So-Brief Hint: Use complementary counting paired with Modulo 3 residue splits across a large localized bounding box to subtract the cases where neither m nor n is a multiple of 3.

2. **(2015 AIME II, Problem 3)** Find the least positive integer N such that N is a multiple of 11, and when N is divided by 2, 3, 4, 5, 6, and 7, the remainders are 1, 2, 3, 4, 5, and 6, respectively.
Answer: 1259
Not-So-Brief Hint: Apply the Symmetrical Chinese Remainder Theorem. Notice that each remainder is $m - 1$, meaning $N \equiv -1 \pmod{m}$ for $m \in \{2, 3, 4, 5, 6, 7\}$. Thus $N + 1$ is a multiple of $\text{lcm}(2, 3, 4, 5, 6, 7) = 420$, so $N = 420k - 1$. Solve the linear congruence modulo 11.

3. **(HMMT February 2025, Algebra & Number Theory, Problem 9)** Let f be the unique polynomial of degree at most 2026 such that for all $n \in \{1, 2, \ldots, 2027\}$, $f(n) = 1$ if n is a perfect square and 0 otherwise. Suppose $\frac{a}{b}$ is the coefficient of x^{2025} in f with $\gcd(a, b) = 1$. Compute the unique

integer $r \in [0, 2026]$ such that $a - rb$ is divisible by 2027.
Answer: 1037
Not-So-Brief Hint: Use Lagrange interpolation over the finite field $\mathbb{F}_p$ for the prime $p = 2027$, translating algebraic coefficient extraction into a summation of quadratic residues modulo 2027.

4. **(1991 USAMO, Problem 3)** Show that for any fixed integer $n \geq 1$, the sequence of exponential powers $2^n, 2^{2^n}, 2^{2^{2^n}}, \ldots$ becomes constant modulo n after a finite number of terms.
Answer: (Proof Framework. You got it.)
Not-So-Brief Hint: Apply strong induction using Euler's Totient Theorem recursively. To evaluate $2^A \pmod{n}$, reduce the tower exponent modulo $\phi(n)$. Because $\phi(n) < n$ for $n > 1$, the chain of moduli strictly decreases until reaching a trivial base case.

5. **(ARML Tournament Archives)** Let x be a positive integer such that $x \leq 100$. Find the total number of valid values of x such that the expression $3x - x^2$ is perfectly divisible by 5.
Answer: 40
Not-So-Brief Hint: Rewrite as a modular equation: $x(3 - x) \equiv 0 \pmod{5}$. By the zero-product property for prime moduli, either $x \equiv 0 \pmod{5}$ or $x \equiv 3 \pmod{5}$. Each interval of 5 numbers contains 2 solutions, giving $20 \times 2 = 40$ values across $[1, 100]$.